2024-25 First Semester
MATH1053 Linear Algebra I

Assignment 1

Due Date: on or before 10:00, 12/Sep/2024 (Thursday), in lecture.

- Write down your **CHN name** and **student number**. Write neatly on **A4-sized** paper (*staple if necessary*) and **show your steps**.
- **Late submissions or answers without steps won't be graded.**

---

1. Let $A = \begin{pmatrix} -2 & 1 & 3 \\ 4 & 1 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 3 & 0 \\ -4 & 0 & 1 \end{pmatrix}$, $C = \begin{pmatrix} 2 & -1 \\ 0 & 1 \\ 2 & 3 \end{pmatrix}$ and $D = \begin{pmatrix} 0 & 1 \\ 2 & 0 \\ -1 & 1 \end{pmatrix}$. Find each of the following matrices.

   (a) $2BC - AD$

   (b) $D^T D$

2. Let $A = \begin{pmatrix} 1 & 0 & -1 & 2 \\ 0 & 3 & 1 & -1 \\ 2 & 4 & 0 & 3 \\ -3 & 1 & -1 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 2 \\ 3 & 1 \\ 0 & -2 \\ 4 & 1 \end{pmatrix}$, and $C = \begin{pmatrix} 3 & -2 & 0 & 5 \\ 1 & 0 & -3 & 4 \end{pmatrix}$.

   (a) Does the matrix $D = ABC$ exist? If so, what is the $(1,2)$-entry d_{12} of D?

   (b) Does the matrix $H = CAB$ exist? If so, what is the $(2,2)$-entry h_{22} of H?

   (c) Does the matrix $J = CBA$ exist? If so, what is the $(1,3)$-entry j_{13} of J?

3. Find conditions on a, b, c, and d such that $AB = BA$ for the matrices below.

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}.$$

4. Let A and B be symmetric $n \times n$ matrices. For each of the following, determine whether the given matrix must be symmetric or could be nonsymmetric:

   (a) $C = A + B$

   (b) $D = A^2$

(c) $F = ABA$

(d) $G = AB + BA$

5. A square matrix A is said to be skew symmetric if $A^T = -A$. Show that if a matrix is skew symmetric, then its diagonal entries must all be 0.

6. Prove that if A is nonsingular, then A^T is nonsingular and $(A^T)^{-1} = (A^{-1})^T$.

7. Show that if A is a symmetric invertible matrix, then A^{-1} is also symmetric.